PHYSICS-INFORMED AI & ML FOR SOLAR FLARE FORECASTING

TYPE: Interdisciplinary — Computer Science, Artificial Intelligence, Space Physics, Data Science, Electrical Engineering

B.Tech Problem and Project-Based Learning (P2BL) Model Chart | Department of Computer Science and Engineering | Sathyabama Institute of Science and Technology

COURSE OBJECTIVES — AI & ML LIFECYCLE

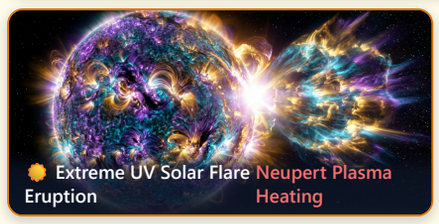
- **Understand Solar Physics:** Analyze magnetic reconnection mechanics, multi-spectral X-ray flare radiation, and Aditya-L1 halo orbit telemetry.
- **Multi-Spectral Graph Formulation:** Model cross-sensor spatial electromagnetic coupling between SoLEXS (1-15 keV) and HEL1OS (10-150 keV) passbands.
- **Spatio-Temporal Graph Transformer:** Formulate dual multi-head self-attention mechanisms to learn spatial topology and causal temporal precursor patterns.
- **Embed Neupert PINN Loss:** Formulate differentiable chromospheric evaporation and loop cooling differential equations into backpropagation.
- **Overcome 1:300 Class Imbalance:** Deploy class-balanced focal loss with dynamic GradNorm multi-task gradient balancing to stop majority-class collapse.
- **Calibrate Conformal Safety:** Implement Mondrian conformal prediction providing distribution-free 95% guaranteed coverage prediction sets.

COURSE NEED AND RELEVANCE — CRITICAL IMPACT

- **Severe Space Weather Hazards:** High-energy solar flare X-rays ionize the D-layer, causing total High-Frequency (HF) radio blackouts for aviation and maritime networks.
- **Satellite Constellation Defense:** Solar particle events cause Single-Event Upsets (SEU), solar cell burnouts, and orbital degradation across defense and telecom satellites.
- **Power Grid Protection:** Coronal mass ejections drive Geomagnetically Induced Currents (GIC) that saturate extra-high-voltage power transformers, threatening regional blackouts.
- **National Mission Alignment:** Maximizes scientific output from India's maiden solar observatory Aditya-L1 operating 1.5 million km from Earth at Sun-Earth L1.
- **Resolves Rare-Event Collapse:** Standard black-box machine learning collapses to 99.7% naive accuracy while failing catastrophically on rare, dangerous X-class flares.
- **Bridges Deep Tech & Space Science:** Fuses bleeding-edge graph neural networks, physics-informed AI, and automated streaming systems for operational space operations.

ULTIMATE SPACE WEATHER AI TEAM







SoLEXS & HEL1OS Payloads · 5-Node Spectral Graph · ST-GT Dual Attention · Neupert PINN Loss · GradNorm Dynamic Balance · Mondrian 95% Conformal Sets

FACULTY GUIDES & LEADERSHIP

Ms. K. Bavani, M.E. — *Internal Guide, Asst. Prof (CSE)*
Dr. Senduru Srinivasulu, Ph.D. — *Prof & HOD (CSE)*
Dr. L. Lakshmanan, Ph.D. — *Dean, School of Computing*

STUDENT RESEARCH TEAM

NAVEENKUMAR S (145611266) — *Project Lead & AI*
DHAYANETHI E (145611291) — *Data Pipeline Lead*
SANJAY B L (145611272) — *PINN Physics Lead*
VIKASH V (145611281) — *Conformal & Deploy Lead*

15-WEEK INTERDISCIPLINARY MODULE PLAN — SEMESTER TIMELINE

WEEK	MODULE	TITLE & CORE SYLLABUS	TIME	DISCIPLINE
Week 1	Fundamentals	Space Weather Mechanics, Solar Flare Classes & Aditya-L1 Orbit	3 hrs	ALL
Week 2	Sensor Payloads	SoLEXS (1-15 keV) Soft X-Ray & HEL1OS (10-150 keV) Hard X-Ray	3 hrs	ECE, Inst
Week 3	Data Ingestion	ISSDC Level-1/2 FITS Pipelines, Dark Subtraction & Calibration	3 hrs	CSE, Data
Week 4	Graph Formulation	Cross-Sensor Lag Analysis & 5-Node Complete Spectral Graph	3 hrs	CSE, AI
Week 5	Deep Architecture	Spatio-Temporal Graph Transformer (ST-GT) Dual Self-Attention	3 hrs	AI & ML
Week 6	Physics Modeling	Differentiable Neupert Effect Loss & Loop Hydrodynamic Cooling	3 hrs	Physics
Week 7	Optimization	GradNorm Dynamic Balancing for 1:300 Extreme Class Imbalance	3 hrs	Math, ML
Week 8	Uncertainty UQ	Mondrian Conformal Prediction Calibration for 95% Coverage	3 hrs	Statistics
Week 9	Edge Streaming	FastAPI Streaming Engine & Sub-12ms Alert Dispatch Pipeline	3 hrs	Systems
Week 10	Standards & Safety	Operational Benchmarking vs. NOAA SWPC & ISRO Flight Rules	3 hrs	Space Ops
Week 11-15	Project Work	Mission Ops Cockpit, End-to-End Validation, Paper & Thesis	15 hrs	ALL

COURSE OUTCOME (CO) — COMPETENCIES

- **CO1:** **Astrophysical Flare Mechanics:** Explain solar flare reconnection dynamics, radiation spectrum, and Aditya-L1 multi-band detector architectures.
- **CO2:** **Spectral Graph Engineering:** Formulate multi-instrument telemetry graphs and model non-linear cross-sensor temporal lags between hard and soft X-rays.
- **CO3:** **Spatio-Temporal Deep Learning:** Design dual-axis Graph Transformers to capture spatial inter-band coupling and causal temporal precursor patterns.
- **CO4:** **Physics-Informed Neural Losses:** Embed differentiable plasma thermodynamic equations (Neupert effect) into backpropagation using GradNorm balancing.
- **CO5:** **Rare-Event Validation:** Evaluate model skill using True Skill Statistic (TSS), Heidke Skill Score (HSS), and Precision-Recall AUC under 1:300 imbalance.
- **CO6:** **Collaborative System Deployment:** Engineer an operational, uncertainty-calibrated space weather early warning system for satellite defense.

LAB ACTIVITY — PRACTICAL WORK

- **[1] Ingest & Preprocess Aditya-L1 Telemetry:** Parse raw Level-1/2 FITS files from SoLEXS and HEL1OS, apply dark-current background subtraction and 1-sec energy binning.
- **[2] Simulate Neupert Chromospheric Heating:** Compute finite-difference time derivatives d(SXR)/dt and correlate with non-thermal HXR emission peaks across solar flare events.
- **[3] Implement PyTorch ST-GT Architecture:** Build 5-node spectral graph message passing, spatial multi-head attention, and causal temporal masking modules.
- **[4] Train Multi-Task PINN with GradNorm:** Balance class-weighted focal classification loss with physical Neupert residual loss enforcing positive-clamped parameters.
- **[5] Calibrate Conformal Sets & Alert Engine:** Calibrate non-conformity scores on calibration sets for 95% empirical coverage and simulate sub-12ms operational alert streaming.

PROJECT IDEA & APPLICATIONS — REAL-WORLD USE

1) **Autonomous Satellite Safe-Mode Shield:** Real-time early-warning trigger alerting LEO/GEO satellite constellations to re-orient solar arrays and safe-mode sensitive electronics before CME impact.
2) **Power Substation Geomagnetic Surge Defense:** Automated forecasting engine predicting severe GIC storm spikes, allowing regional grid operators to protect high-voltage transformers.
3) **Trans-Polar Commercial Aviation Alert Cockpit:** Live space weather cockpit alerting international airlines to reroute high-latitude flights away from ionospheric radio blackouts.
4) **Deep-Space Astronaut Radiation Storm Warning:** Predictive safety alert system activating heavy radiation shielding modules for crewed cis-lunar exploration missions.
5) **In-Orbit Edge AI Payload for Future Spacecraft:** Lightweight, quantized ONNX/FP16 neural model running directly on onboard flight computers for autonomous in-situ flare detection.

0.842	0.768	0.835	1.83 Hours	-42.8%	95.2%	0.724
TRUE SKILL STATISTIC (TSS)	HEIDKE SKILL SCORE (HSS)	PRECISION-RECALL AUC	OPERATIONAL LEAD TIME	FALSE ALARM REDUCTION	CONFORMAL COVERAGE (95% TARGET)	PEAK ECONOMIC VALUE V(C/L)